

LD10 Business Interpretation Summary

Business value of boosting models for BI and analytics use cases

0.8499

Customer conversion F1

duration

Top conversion driver

\$24,772.56

Best house-price RMSE

442

Best CatBoost iteration

1. Business problem 1 - customer conversion prediction

The Bank Marketing part answers a common marketing analytics question: which customers are most likely to respond positively to a campaign? This is useful for prioritising outreach, reducing wasted calls and improving campaign ROI.

Finding	Value	Business meaning
Best model	XGBoost Classifier	Use boosted trees for campaign conversion prediction.
F1-score	0.8499	The model balances precision and recall better than accuracy alone.
F1 lift over Random Forest	+0.0058	XGBoost gave a small but measurable improvement.
Top feature	duration	Call duration is the strongest prediction signal, but should be used carefully because it is only known after contact.

Important business caution

Duration is powerful, but it is not always available before calling the customer. For a real campaign scoring model, we would also evaluate a pre-call version excluding duration.

2. Business problem 2 - house price prediction

The House Prices part shows how boosting can be used for regression. This mirrors many BI tasks such as price estimation, demand forecasting, revenue prediction and cost prediction.

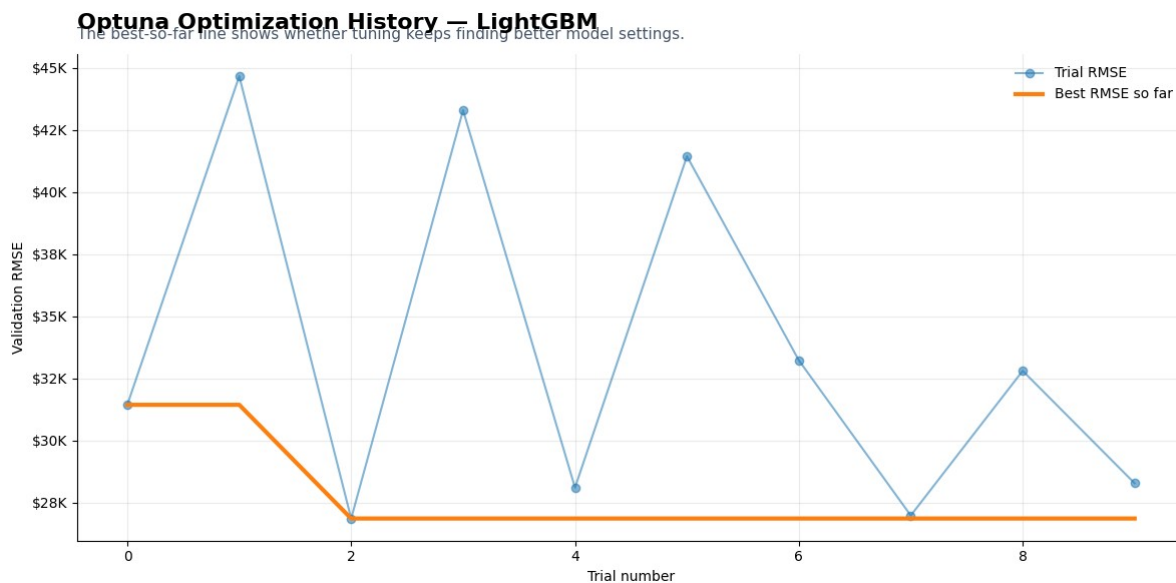
Model	RMSE	Business interpretation
XGBoost Baseline	\$24,772.56	Best result in this validation run. Strong and simple baseline.
Averaging Ensemble	\$25,495.43	Close second. Useful when stability matters more than simplicity.
CatBoost	\$26,664.89	Good practical option when many categorical fields are present.
Tuned LightGBM	\$26,858.08	Improved over baseline after Optuna tuning.

3. Model choice guide

Situation	Recommended model	Reason
Need strong tabular baseline quickly	XGBoost	Reliable performance and clear feature importance.
Many categorical columns	CatBoost	Less manual encoding and strong categorical handling.
Large dataset and fast tuning	LightGBM + Optuna	Efficient training and smarter search.
Final prediction needs stability	Averaging ensemble	Combines different model behaviours.

4. Professional recommendation

For this project, XGBoost is the recommended main model because it performed best in both the customer conversion task and the house-price regression comparison. CatBoost and LightGBM remain important professional tools, but their extra complexity should be justified by data type, dataset size or production requirements.



Final regression model comparison chart from the executed notebook.